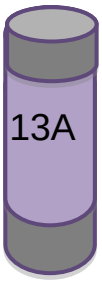
In order to work out what size fuse is needed for an appliance, you need to know the current the appliance will normally use. Most appliances have a label on them to inform you what the power rating is.



or



or



To work out the current use the following formula:

power = current x potential difference

$P = I \times V$ or $I = P/V$

P is in watts
I is in amps
V is in volts

Task

Complete the table below.
Decide which fuse, 3A, 5A or 13A would be appropriate. Remember that the fuse size needs to be slightly higher than the normal current.
Find the power rating of three more appliances and complete the table.

	Appliance	Power rating (watts)	Potential difference (volts)	Current (amps)	Fuse needed
1	toaster	1500	240	6.25	13
2	microwave	900			
3	washing machine	500			
4	kettle	3 kW			
5	laptop	60			
6	iron	1 kW			
7	TV recorder	25			
8					
9					

10					

Answers

	Appliance	Power rating (watts)	Potential difference (volts)	Current (amps)	Fuse needed
1	toaster	1500	240	6.25	13
2	microwave	900	240	3.75	5
3	washing machine	500	240	2.08	3
4	kettle	3 kW	240	12.5	13
5	laptop	60	240	0.25	3
6	iron	1 kW	240	4.17	5
7	TV recorder	25	240	0.10	3
8					
9					
10					

Answers

	Appliance	Power rating (watts)	Potential difference (volts)	Current (amps)	Fuse needed
1	toaster	1500	240	6.25	13
2	microwave	900	240	3.75	5
3	washing machine	500	240	2.08	3
4	kettle	3 kW	240	12.5	13
5	laptop	60	240	0.25	3
6	iron	1 kW	240	4.17	5
7	TV recorder	25	240	0.10	3
8					
9					
10					